Homework 3

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```r
library(astsa)
```

I.

2.10)

We have $x_t = \mu + w_t + \theta w_{t-1}, \quad w_t \sim wn(0, \sigma_w^2)$.

a)

$$E[x_t] = E[\mu + w_t + \theta w_{t-1}] = \mu + 0 + 0 = \mu$$

b)

$$\gamma_x(0) = Var(x_t) = Var(\mu + w_t + \theta w_{t-1}) = Var(w_t) + \theta^2 Var(w_{t-1}) = \sigma_w^2(1 + \theta^2)$$

$$\gamma_x(1) = Cov(x_{t+1}, x_t) = Cov(\mu + w_{t+1} + \theta w_t, \mu + w_t + \theta w_{t-1}) = Cov(\theta w_t, w_t) = \sigma_w^2 \theta$$

$$\gamma_x(-1) = Cov(x_{t-1}, x_t) = Cov(\mu + w_{t-1} + \theta w_{t-2}, \mu + w + \theta w_{t-1}) = Cov(w_{t-1}, \theta w_{t-1}) = \sigma_w^2 \theta$$

c)

For x_t to be stationary expected value needs to be constant and not depend on t. The Auto-Covariance function needs to depend only on the lag $h = |s - t|$ and not the time s or t. We can see both of these are met in the last two results. For x_t to be stationary for all $\theta \in \mathbb{R}$ we simply need to show that $\gamma_x(0) = Var(x)$ is always non negative. Since $(1+\theta^2)$ is always non negative, we can say that x_t is stationary for all $\theta \in \mathbb{R}$.

d)

We have $Var(\bar{x}) = \gamma_x(0)/n$ since x_t is white noise.

i)

$$Var(\bar{x}) = \gamma_x(0)/n = 2\sigma_w^2/n$$

ii)

$$Var(\bar{x}) = \gamma_x(0)/n = \sigma_w^2/n$$

iii)

$$Var(\bar{x}) = \gamma_x(0)/n = 2\sigma_w^2/n$$

e)

The accuracy in the estimate of μ is best when $\theta = 0$, the variance of our estimate is 1/2 that of $\theta = 1$ and $\theta = -1$.

2.11)

a)

We are generating the series $x_t = w_t$ where $w_t \sim wn(0, \sigma_w^2)$.

```
set.seed(3)
w = rnorm(500)
acf(w, lag.max = 20, plot = FALSE)
```

```
Autocorrelations of series 'w', by lag

     0      1      2      3      4      5      6      7      8      9     10 
 1.000 -0.013 -0.025  0.028  0.022 -0.011  0.072  0.016 -0.010 -0.039 -0.007 
    11     12     13     14     15     16     17     18     19     20 
 0.004 -0.057  0.025 -0.106 -0.109 -0.004 -0.013 -0.079 -0.026 -0.003 
```

We find the actual ACF:

$$\gamma_x(0) = Var(x_t) = \sigma_w^2$$

$$\gamma_x(1) = Cov(x_t, x_{t-1}) = Cov(w_t, w_{t-1}) = 0$$

So:

$$\gamma_x(h) = \begin{cases} \sigma_w^2 & h = 0 \\ 0 & h \geq 1 \end{cases}$$

And then the actual ACF is:

$$\rho(h) = \begin{cases} 1 & h = 0 \\ 0 & h \geq 1 \end{cases}$$

We can see that the sample ACF varies slightly around the true value 0.

b)

```
set.seed(4)
w_2 = rnorm(50)
acf(w_2, lag.max = 20, plot = FALSE)
```

```
Autocorrelations of series 'w_2', by lag

     0      1      2      3      4      5      6      7      8      9     10
 1.000  0.003 -0.047 -0.074  0.087  0.133  0.187 -0.025  0.062 -0.217 -0.029
    11     12     13     14     15     16     17     18     19     20
 0.255 -0.050  0.041  0.092  0.012 -0.019  0.013  0.081  0.029 -0.076
```

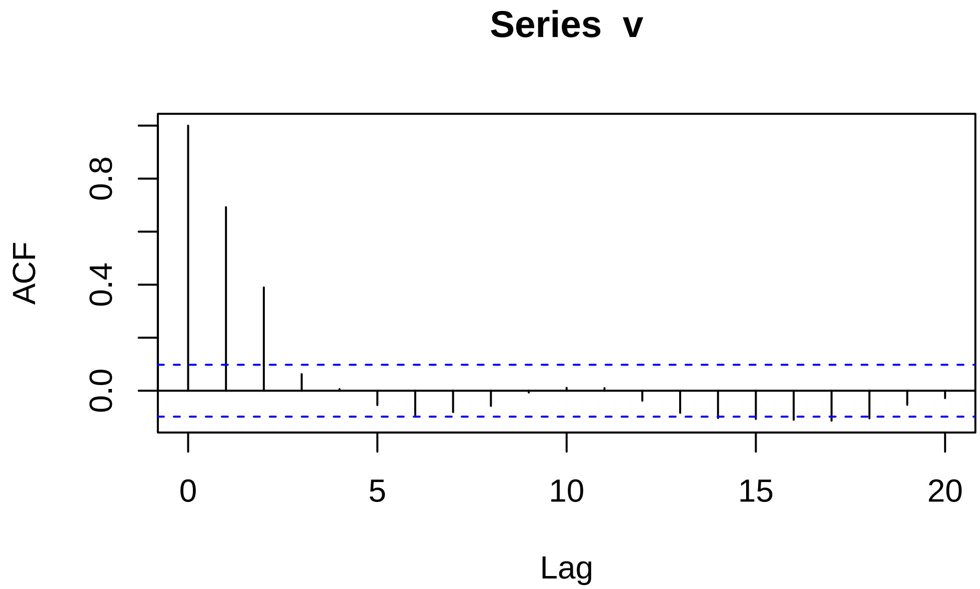
Now the estimates of the ACF vary more than with $n = 500$.

2.12)

We are generating the series $v_t = \frac{1}{3}(w_{t-1} + w_t + w_{t+1})$ where $w_t \sim wn(0, \sigma_w^2)$.

```
set.seed(11)
w = rnorm(500+51)
```

```r
v = filter(w, sides = 2, filter = rep(1/3, 3))[100:500]
acf(v, lag.max = 20, plot = TRUE)
```




```r
acf(v, lag.max = 20, plot = FALSE)
```

```
Autocorrelations of series 'v', by lag

     0      1      2      3      4      5      6      7      8      9     10 
 1.000  0.692  0.390  0.063  0.006 -0.054 -0.092 -0.081 -0.057 -0.007  0.011 
    11     12     13     14     15     16     17     18     19     20 
 0.010 -0.038 -0.084 -0.103 -0.106 -0.110 -0.113 -0.104 -0.053 -0.028 
```

The actual ACF is as follows:

$$\gamma_v(0) = Var(v_t) = \frac{1}{3}\sigma_w^2$$

$$\gamma_v(1) = Cov(v_{t+1}, v_t) = Cov(\frac{1}{3}(w_t + w_{t+1} + w_{t+2}), \frac{1}{3}(w_{t-1} + w_t + w_{t+1}) = \frac{2}{9}\sigma_w^2$$

$$\gamma_v(2) = Cov(v_{t+2}, v_t) = Cov(\frac{1}{3}(w_{t+1} + w_{t+2} + w_{t+3}), \frac{1}{3}(w_{t-1} + w_t + w_{t+1}) = \frac{1}{9}\sigma_w^2$$

So we have:

$$\gamma_v(h) = \begin{cases} \frac{3}{9}\sigma_w^2 & h = 0 \\ \frac{2}{9}\sigma_w^2 & h = 1 \\ \frac{1}{9}\sigma_w^2 & h = 2 \\ 0 & h \geq 3 \end{cases}$$

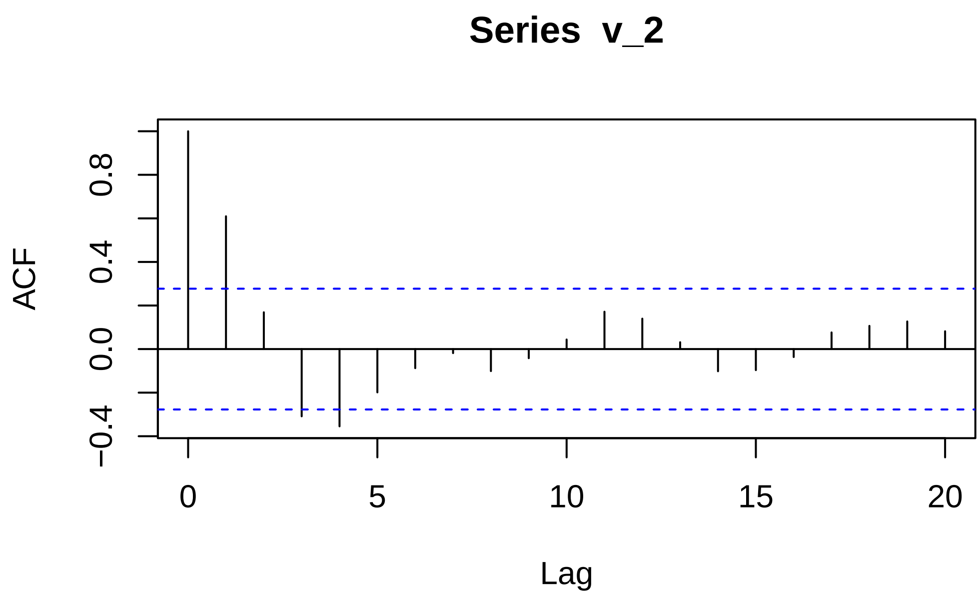
And the ACF is as follows:

$$\rho_v(h) = \begin{cases} 1 & h = 0 \\ \frac{2}{3} & h = 1 \\ \frac{1}{3} & h = 2 \\ 0 & h \geq 3 \end{cases}$$

The sample ACF is pretty close to the true values.

b)

```
set.seed(15)
w_2 = rnorm(50+51)
v_2 = filter(w, sides = 2, filter = rep(1/3, 3))[2:51]
acf(v_2, lag.max = 20, plot = TRUE)
```




```
acf(v_2, lag.max = 20, plot = FALSE)
```

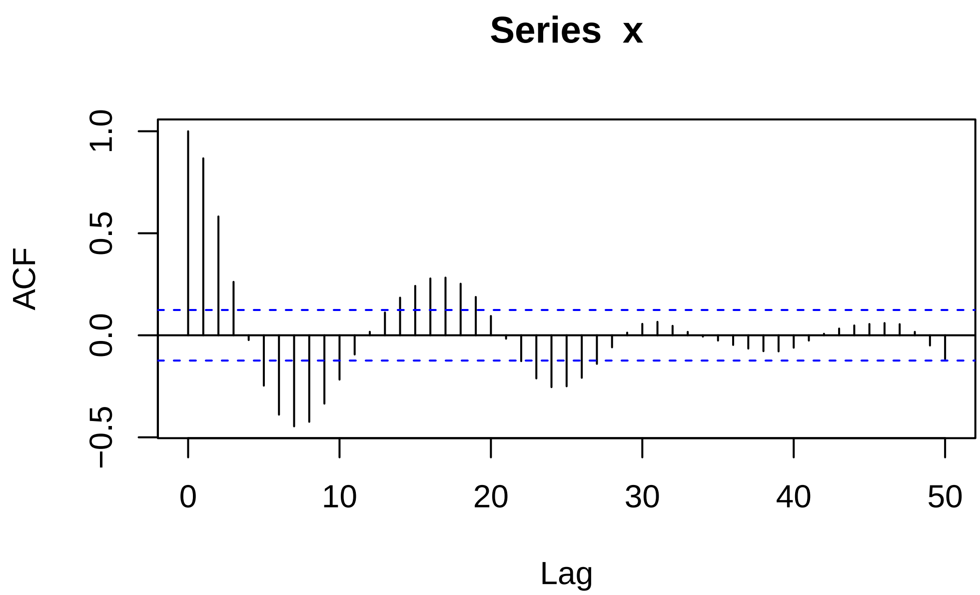
```
Autocorrelations of series 'v_2', by lag

     0      1      2      3      4      5      6      7      8      9     10
 1.000  0.610  0.169 -0.309 -0.355 -0.199 -0.088 -0.019 -0.101 -0.042  0.044
    11     12     13     14     15     16     17     18     19     20
 0.172  0.140  0.032 -0.102 -0.097 -0.037  0.076  0.107  0.127  0.082
```

We can see that the estimates start to vary from the true values by quite a bit, lag 2 is .169 which is quite far from the true 1/3, and lags 3 and 4 are both far from 0. By reducing n we have increased the variance in our estimator.

2.13)

```
set.seed(1000)
w = rnorm(250 + 50)
x = filter(w, filter = c(1.5, -.75), method = "recursive")[-(1:50)]
acf(x, lag.max = 50, plot = TRUE)
```

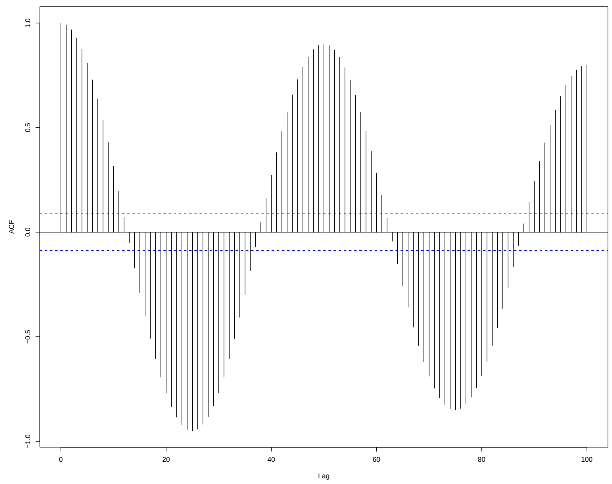


We can see a strong negative correlated cycle starting at lag 4, that alternates to positive correlation and back to negative correlation about every 8 observations. The structure gradually fades to near zero as observations become more distant.

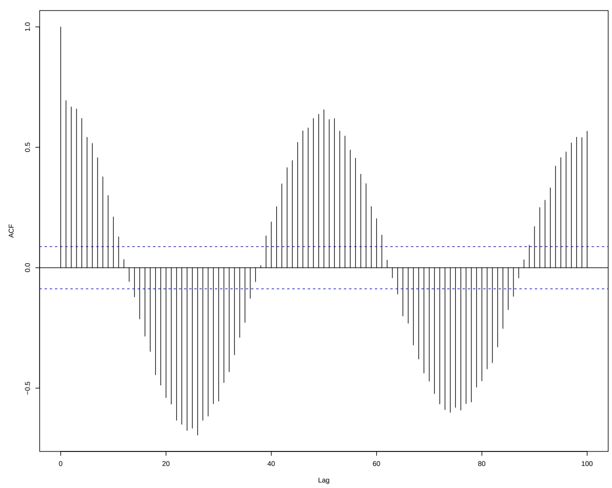
2.14)

```
t = 1:500
cs = 2*cos(2*pi*(t+15)/50)
w = rnorm(500)
par(mfrow = c(3,1))
acf(cs,lag.max = 100, plot = TRUE)
acf(cs + w,lag.max = 100, plot = TRUE)
acf(cs + 5*w,lag.max = 100, plot = TRUE)
```

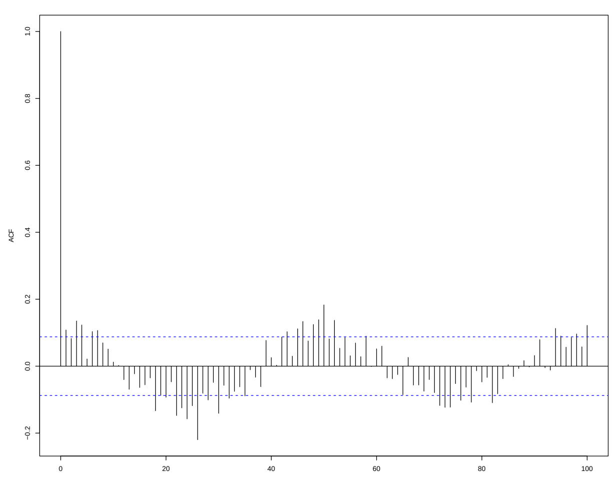
Series cs



Series cs + w



Series cs + 5 * w



We can see that as the white noise component becomes larger the magnitude of the cycle becomes smaller in the ACF.

II.

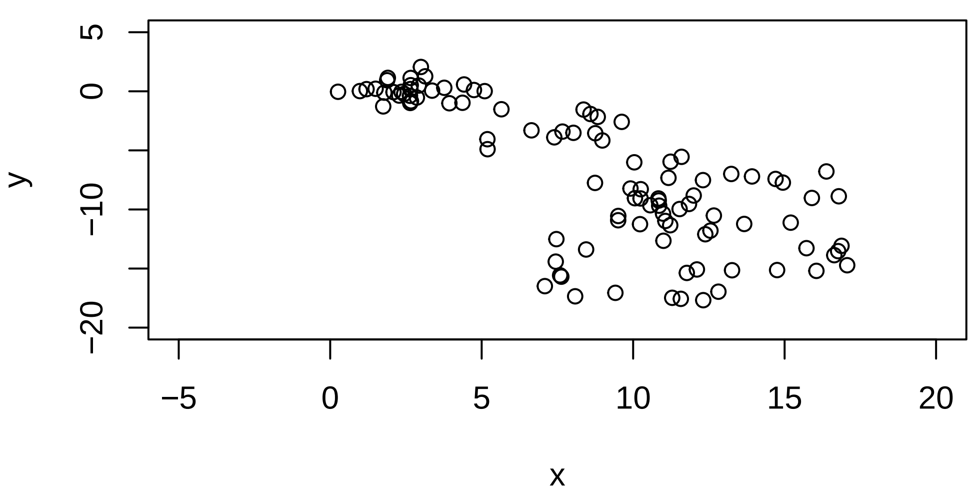
a)

```
set.seed(15)
w_1 = rnorm(100)
w_2 = rnorm(100)

x = cumsum(w_1)
y = cumsum(w_2)
```

i)

```
plot(x,y, xlim = c(-5, 20), ylim = c(-20, 5))
```



There appears to be a pattern of negative correlation between x and y.

ii)

I believe H_0 would be rejected in this case, the scatter plot shows a strong negative association between x and y. This will lead to the estimate $\hat{\beta_1}$ being quite large.

In general though, I would expect $H_0 : \beta_1 = 0$ to not be rejected because x and y have no actual relationship with each other.

iii)

```
model_a = lm(y~x)
summary(model_a)$coefficients[2,]
```

```
     Estimate     Std. Error        t value       Pr(>|t|)
-9.614760e-01  8.554153e-02 -1.123987e+01  2.527813e-19
```

We can see that the null hypothesis of $\beta_1 = 0$ is rejected at $\alpha = .05$.

b)

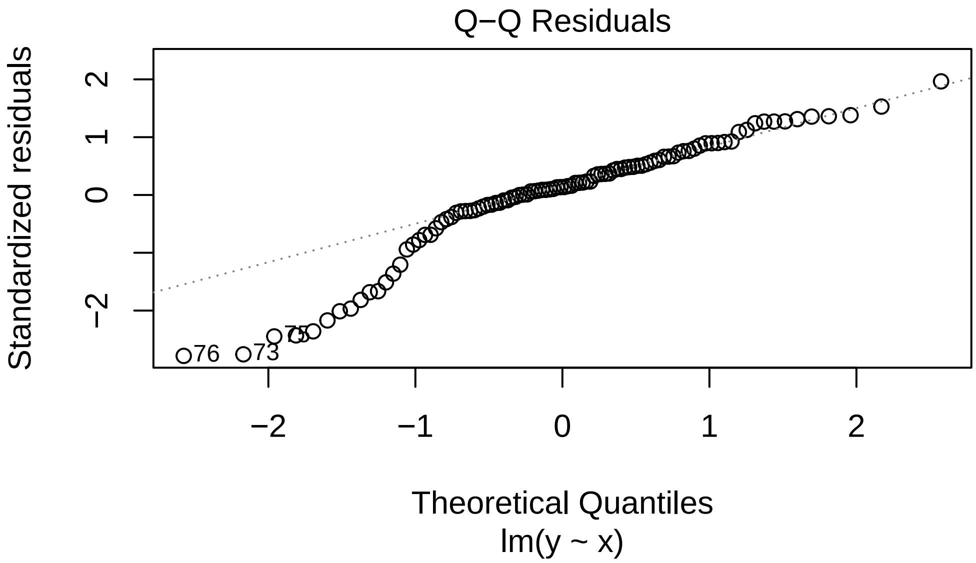
```
set.seed(50)
reject = rep(0,1000)
for(i in 1:1000){
  w_1 = rnorm(100)
  w_2 = rnorm(100)
  x = cumsum(w_1)
  y = cumsum(w_2)
  model = lm(y~x)
  p_value = summary(model)$coefficients[2,4]
  ifelse(p_value < .05, reject[i] <-  "Rejected", reject[i] <-  "Not Rejected")
}
table(reject)
```

```
reject
Not Rejected     Rejected
         248          752
```

We can see that $H_0 : \beta_1 = 0$ is rejected 752 of the 1000 trials, this is surprising to me. I expected H_0 to be rejected around the expected 5% of times with this many trials since we know there is no relationship between x and y.

One of the assumptions that is placed upon a linear regression model is that that e_i is i.i.d. normal. If we take a look at the qq plot we can see that assumption is violated.

```r
plot(model_a, which = 2)
```



Since we have the linear model:

$$y_i = \beta_0 + \beta_1 x_i + e_i$$

We can solve for e_i:

$$e_i = y_i - \beta_0 - \beta_1 x_i$$

We know that $y_i \sim N(\sum_{l=1}^{i-1} y_l, \sigma_w^2)$:

So, the true distribution of e_i is $N(\sum_{l=1}^{i-1} y_l - \beta_0 - \beta_1 x_i, \sigma_w^2)$ which shows that the distribution of e_i changes as i changes, violating the assumption of i.i.d errors.

III.

1)

a,c,d

2)

b

3)

d

4)

d

5)

b

6)

c

7)

b/a (the expend feature falls far outside the range in our sample of 3.656 to 9.774, we can compute the interval but we shouldn't draw anything from it. The computed interval is a.)

8)

a

9)

e

10)

d